

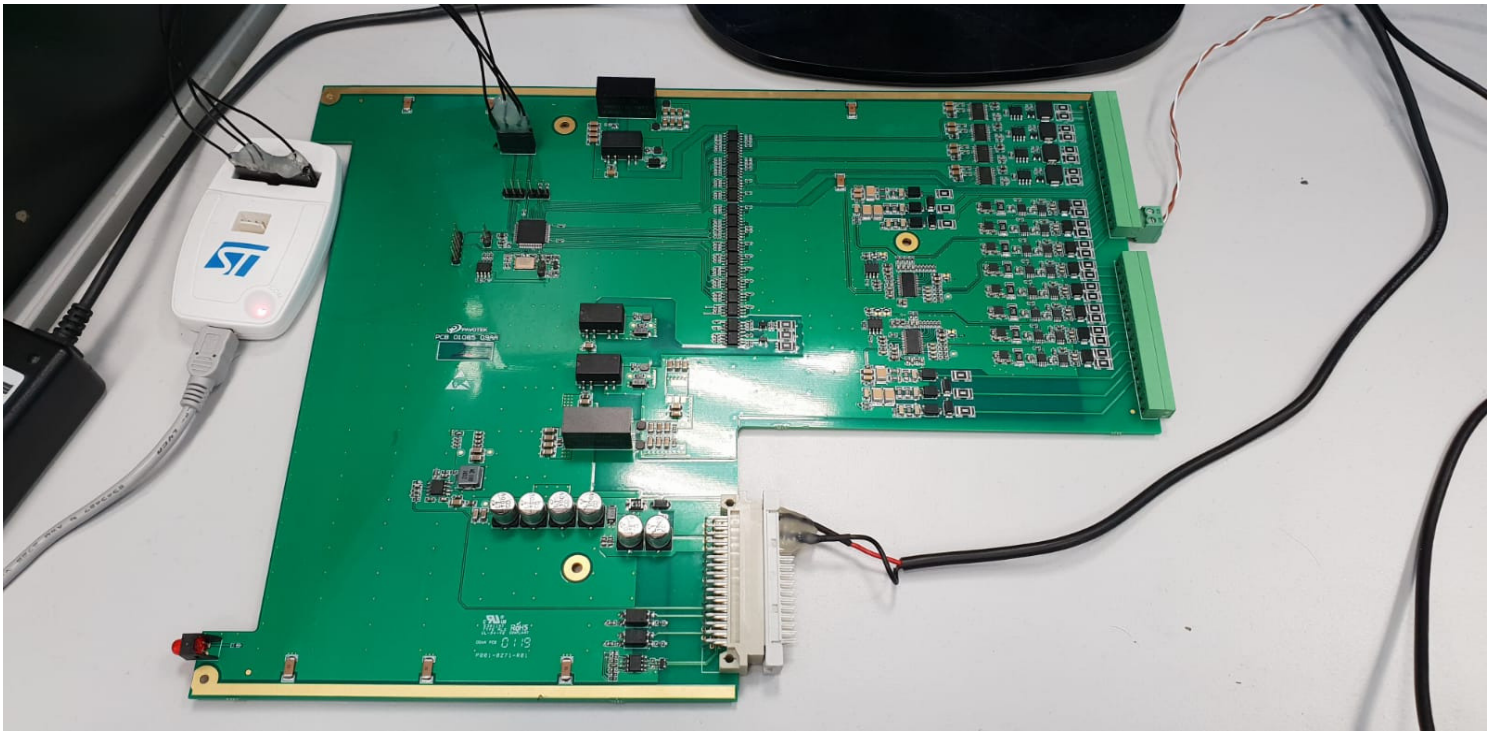
READING ANALOG SIGNALS AND CONVERTING TO DIGITAL DATA WITH PROGRAMMING 16-BIT ADC USING ARM BASED MCU

INTRODUCTION

Il have done my training program at PAVOTEK A.Ş. which is a company that mainly produces electronic system designs for Turkish Defence Industry. PAVOTEK has five main divisions which are energy, Power, avionics, communication, IoT electronics,etc.... The division that I worked for was energy division eventually. At the beginning, our supervisor assigned me the adc programming topic to research. Then I was given a project which aimed to program 16 bit ADC with ARM based MCU for precision sensor measurement. On the other hand, we have done lots of tests on ADC units. During my training program, I have participated in these tests and take down some notes by observing the input and output data during the testing process.

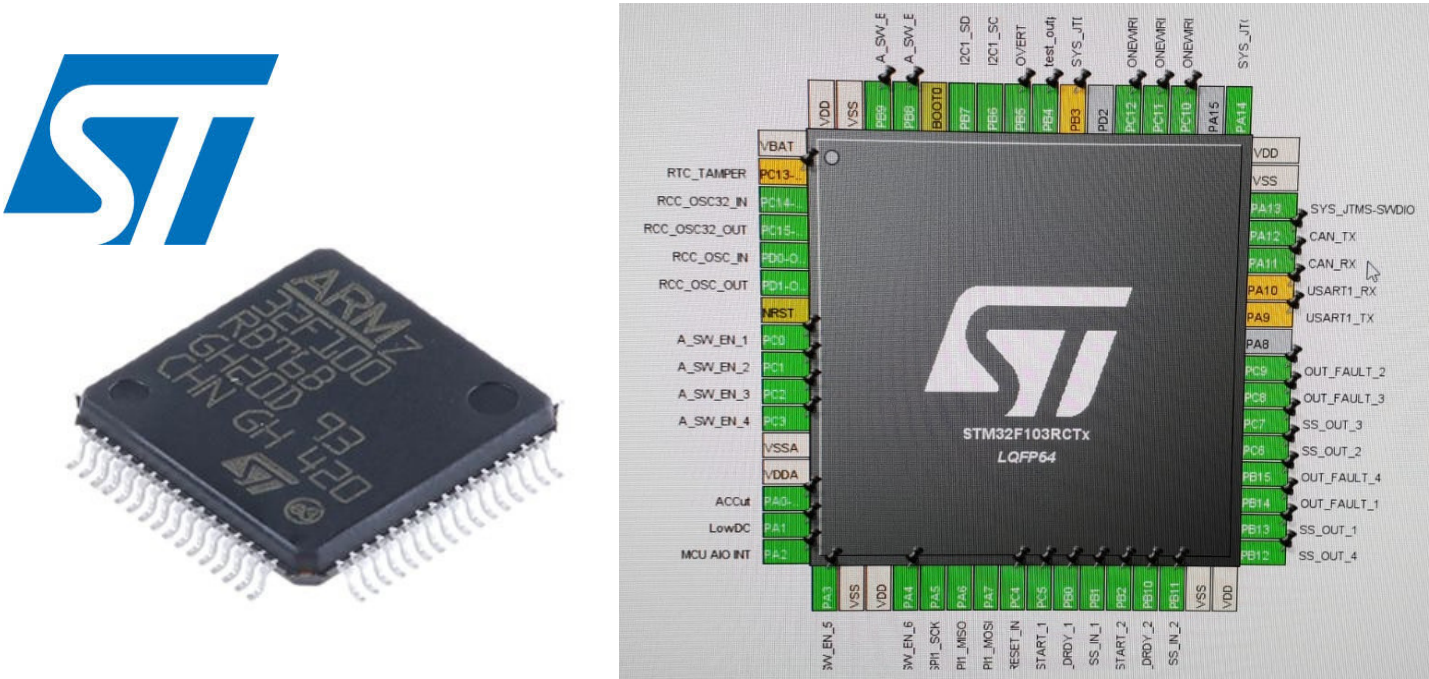
PURPOSE

My internship project is “Reading Analog Signals and Converting to Digital Data with Programming 16-Bit ADC Using ARM Based MCU”. In this project we aim programming 16-Bit ADS1148 ADC to read analog signals and convert them to digital data with using STM32F103RCTx MCU. ADS1148 will accept clock signals and differential analog inputs and output digital data. Clock signals will be produced by an external oscillator and differential inputs will be selected by the internal multiplexer. Programming of ADS1148 will be done using SPI communication with STM32F103RCTx MCU in C language. Atollic TrueSTUDIO and STM32CubeMX programs will be used in code development process. At the end, We want to read meaningful digital data which can be used for precision analog sensor measurement.



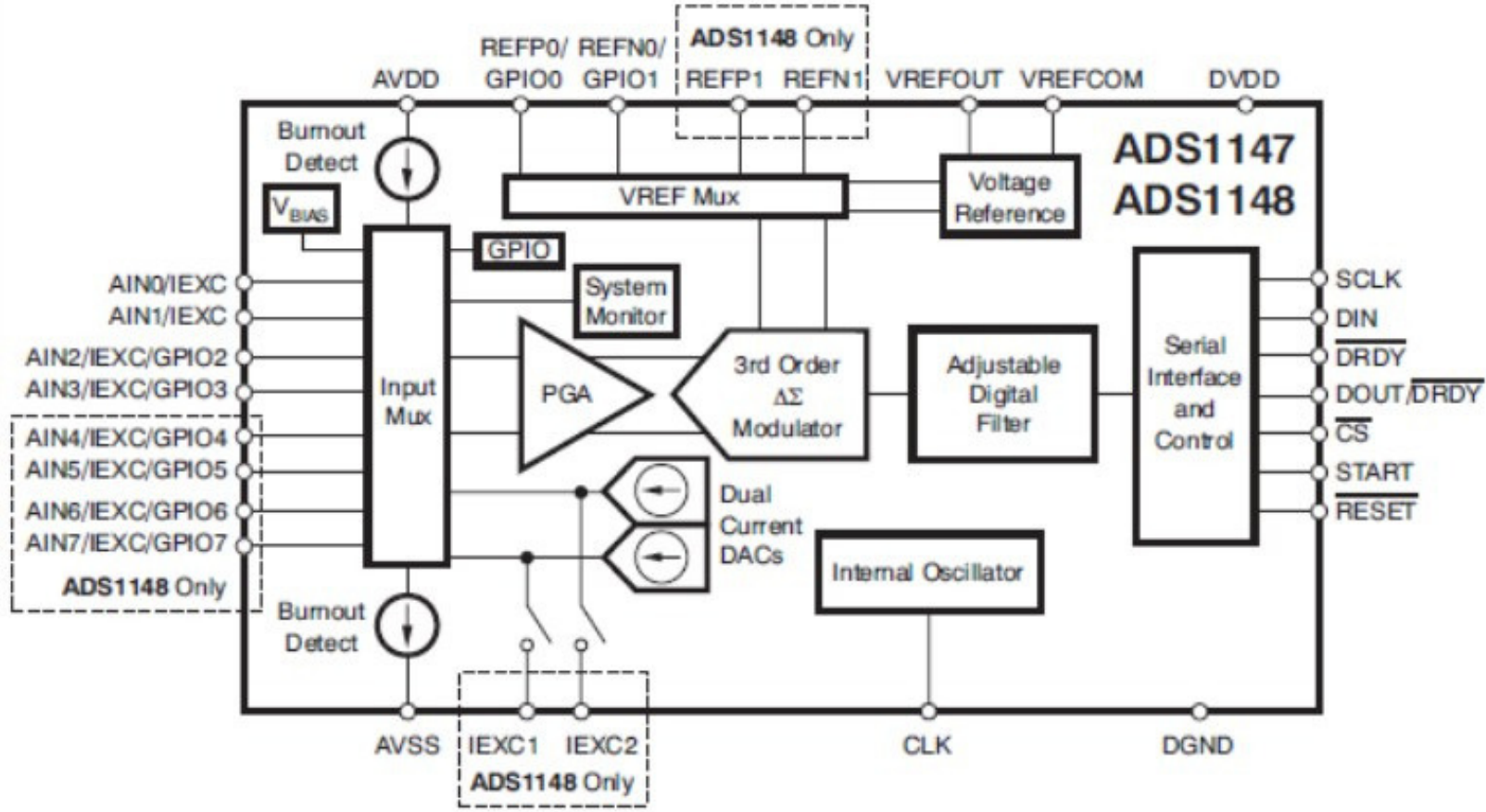
METHODS & MATERIALS

My internship project is “Reading Analog Signals and Converting to Digital Data with Programming 16-Bit ADC Using ARM Based MCU”. In this project we aim programming 16-Bit ADS1148 ADC to read analog signals and convert them to digital data with using STM32F103RCTx MCU. ADS1148 will accept clock signals and differential analog inputs and output digital data. Clock signals will be produced by an external oscillator and differential inputs will be selected by the internal multiplexer. Programming of ADS1148 will be done using SPI communication with STM32F103RCTx MCU in C language. Atollic TrueSTUDIO and STM32CubeMX programs will be used in code development process. At the end, We want to read meaningful digital data which can be used for precision analog sensor measurement.



CONCLUSION

In university we are mainly focused on the theoretical parts but in the business life the most important thing is to implement what you I earned so far to the real life problems. In my internship program I mostly researched and learned about ADC, MCUs, relays, embedded programming with C language. My aim was to program an adc to convert analog input voltages to digital and show the data. Luckily I had really good supervisors and resources to complete the project that I’m assigned with.



RESULTS & DISCUSSION

During my attempts to run my code on the board and read the data correctly I faced lots of problems.

1-The first problem was I couldn’t read any data when I try to readout adc output in 2’s complement format from adc registers. When I read the data I always get 0’s.While trying to understand the cause of the problem I found that the differential input pin that needs to get 2.5 V(which we use as reference) was 0.8 and it was caused by an hardware error. After we fixed this error and test our software again, nothing changed we still got 0’s from the register that we read. Thus our company supervisor advised me to re-read the ADS1148 adc’s datasheet. I found out that input range of the adc was [-vref/gain, vref/gain]with my gain equals to 1, my input range was [-vref, vref]. Then suddenly I realised that we weren’t using vref pins to provide a reference voltage, which means my vref0 pin(by default) was 0 V. The reason for that was our thinking that it will be enough to give the reference voltage(2.5V) from the differential input, but actually it was not enough. The adc was only checking the vref pins for reference voltage thus our reference voltage was being read as 0 from vref0 pin. When we changed it to 2.5 we read meaningful adc output in 2’s complement from the registers. That meant we have changed our input range to correct interval and we were reading digital data.

2-On the other hand I needed the convert 2’s complement data that we read to decimal value,add 2^15 to it to get an absolute binary value in our adc graph. After doing that I multiplied this absolute binary value with step size to get the value that is coming to the ADC input in digital format. That was the voltage which was manipulated by opamps. To get the the real input from the voltage supply I subtracted 2.5V from the previous result so I got ADC Digital Input Voltage and lastly to get the voltage offered from the supply we multiply ADC Digital Input Voltage with -5 and add 1. So I got Power Supply Input Digital Voltage which was what I want. After all the conversion I was able to correctly convert 5-10V analog input to digital value .

3-My aim was to be able to convert 0-10V analog input to digital so I tried to understand why I can only read 5-10V correctly. After research I found that in hardware that was an other problem.One of the pins of INA159 opamp was connected to false value and after I changed it to 5V the problem solved. I had been able to read and convert 1.2-10V analog data to digital.

In conclusion, I can say that I successfully reached the goal of the Project except I was reading data in 1.2-10V interval instead of 0-10V . (If I changed the reference voltage to a higher voltage I could read in 0-10V interval but my internship ended.) At the end we were able to convert 1.2-10V analog input voltage to digital and see the result.